

# 11 — Liquid Track: Architecture and Materials

## v1.2 — hardware, materials law, and operating modes

**Function:** Buildable architecture for doc 10: the film-cell bank, aerosol control, sealed still, ERV and CO<sub>2</sub>-battery placements, brine circulation, thermal bus, raw-water circuit, and the bubble-column annex. Shared doctrines and the safety register → doc 00.

### 1. The two-worlds rule (load-bearing)

Everything on the brine/raw-water side is **metal-free**: plastics, elastomers, titanium. Everything on the hot-water/glycol side is ordinary hydronics (copper, brass, steel welcome). **Titanium heat exchangers are the only membrane between the worlds** — never let the fluids mix. These rules are **never relaxed for land installs** — the desiccant itself is the chloride source (doc 00 §1).

#### Master materials table (every brine- or raw-water-wetted component)

| Status                     | Materials                                                                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use freely                 | PVC (≤60 °C), CPVC (≤93 °C), PP, HDPE/PE/PE-RT, PEX, PVDF, titanium, EPDM, FKM, silicone, ceramic/alumina, glass, acetal (POM)                                                                                                      |
| Never                      | Copper, brass, bronze; aluminum (incl. anodized); zinc/galvanized; mild steel; 304 SS; <b>nylon/polyamide</b> (CaCl <sub>2</sub> is a documented stress-cracking agent for loaded PA parts — the automotive road-salt failure mode) |
| Marginal — don't design in | 316/316L: brief splash only; never warm immersion or crevice geometry (threads, brazed channels, gasket lands)                                                                                                                      |

Wetted fasteners: **PP / PVC / PVDF only**. Rule of thumb: parts sold for pools, aquaculture, or chemical dosing are probably right; parts sold for plumbing hide brass. **Any brazed-plate exchanger (304 + copper braze) is forbidden on brine or raw water** — glycol  buffer isolation duty only.

Salt: plain 94–97% CaCl<sub>2</sub> pellet or 83–87% flake — SDS-check for no additives (no ferrocyanide, dye, or MgCl<sub>2</sub> blend). Add salt to water slowly (strongly exothermic). Tap water is fine.

### 2. Primary absorber — modular film-cell bank

Built entirely from cooling-tower commodity parts (cross-fluted film fill, matched distribution pads, louvers, PVC chevron drift eliminators).

#### Cell (the design quantum):

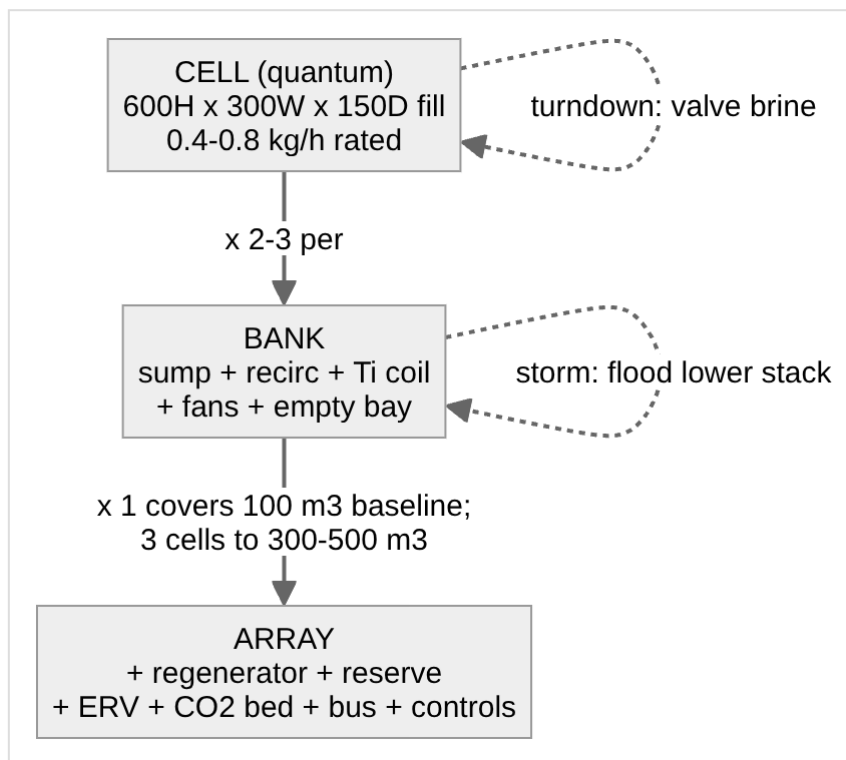
| Element                 | Spec                                                                                                                                                    |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fill envelope           | 600 H × 300 W × 150 D mm; cross-flow, 300 mm air path                                                                                                   |
| Between stacks          | Redistribution tray + drip header — resets maldistribution every 300 mm (tilt discipline; marine)                                                       |
| Brine feed              | Drilled-PVC drip header + 100 mm distribution pad; isolation valve per cell                                                                             |
| Air side                | Inlet louver, outlet chevron drift eliminator, backdraft damper                                                                                         |
| Rated capacity          | <b>0.4–0.8 kg/h water at 0.6–1 m/s face velocity</b> (band PENDING test I, which now includes the ~123 m <sup>3</sup> /h mixed-mode total-flow point)   |
| Turndown                | To near-zero: valve off brine (dry fill ≈ 20–40 Pa) or stop fan                                                                                         |
| Storm fallback (marine) | <b>Flooded mode</b> — raise sump level to submerge the lower stack; ΔP 100–300 Pa; tilt tolerance approaches pool-class. A level setpoint, not a gimbal |

**Bank** = HDPE sump + PP mag-drive recirculation pump (15–30 W) + Ti cooling coil

- 2–3 cell bays + 150 mm EC duct fans (25–40 W each) + **one deliberately empty bay**. Mixed-mode duty (0.88 kg/h peak) sits inside a 2-cell bank at mid-rating.

**The binding constraint is wetting, not airflow.** Desiccant surface tension demands **~7–11 L/min irrigation per cell** (150–240 L/min·m<sup>2</sup>); underwetting collapses mass transfer nonlinearly. The film contactor is a *high-recirculation* device (~50 passes per unit concentration change); only a slow **peristaltic bleed (0.04–0.35 L/min)** exchanges with the regenerator. The humidity floor is set by **sump concentration**, not per-pass approach.

**Contactor depth (NTU sizing, pre-test):** 85% approach → NTU ≈ 1.9; with the literature K·a spread (1–3 kg/m<sup>3</sup>·s) the central estimate is **two 300 mm stages in series**, conservative three. Whole cabinet 0.25 × 0.25 × 1.1 m — slim-locker / closet envelope. Outlet RH at 1/2/3 stages is the NTU/m measurement (test I).



### 3. Aerosol control (gates cabin connection — safety register item 3)

Salt aerosol in breathed air is a corrosion and respiratory hazard. Two commodity stages in every mode: **PVC chevron drift eliminator** at the media face (drift 0.001–0.005% of circulating liquid) then **PP mesh demister** before the supply duct. No metal mesh of any grade.

**Mandatory acceptance test B:** a bare mild-steel coupon downstream for a week at highest design face velocity, trickle *and* flooded — any rust means redesign before any cabin connection.

**Two distinct duties — do not conflate.** The chain above is *outlet-side*: it removes  $\text{CaCl}_2$  mist generated by the contactor itself — a fine, self-generated droplet population at face velocity — and it is gated by test B. The *intake-side* duty is different: removing ambient sea-salt loading before air enters the system, met by the established marine train specified in doc 22 §3 (multi-stage inertial vane separator → coalescing stage → final filter, drained sump) under the same positive-pressure envelope condition. An intake train does **not** substitute for the outlet chain and does **not** relax test B; conversely the outlet chain does not protect the ERV or any downstream metal from ambient chloride. And under the two-worlds law (§1) an intake train never relaxes the chloride materials rules on this track: the desiccant is itself the chloride source regardless of site.

### 4. Regenerator — sealed still with a graceful-degradation ladder

Primary mode: **sealed still**. Shallow HDPE/PP/CPVC tray (0.3–0.5  $\text{m}^2$  pool), sealed lid sloped  $\geq 15^\circ$ , condensate gutter, anti-entrainment PP baffle; verify condensate TDS  $\approx 0$  (test G). At 60–93 °C the brine's vapor pressure (14–70 kPa) dwarfs the sink-cooled condenser's (~4–5 kPa); the full duty leaves as **distillate** with no air stream, no demister, no salty exhaust — the X8 doctrine embodied. Rate scales with pool + condenser area (~1–2  $\text{kg/h}\cdot\text{m}^2$  at design  $\Delta T$ ).

**Temperature is a rate lever, not an equilibrium wall** (finding X2 — the still never hits the solid track's zero-driving-force floor):

| Pool temp    | Cheapest legal material         | Still rate vs 85 °C | Notes                                         |
|--------------|---------------------------------|---------------------|-----------------------------------------------|
| 93 °C        | CPVC                            | ~1.45×              | CPVC's only unique purchase                   |
| <b>85 °C</b> | <b>PP or CPVC</b>               | <b>1.0</b>          | Heat-rich: 43–44 wt% → 7.5–9 g/kg floor       |
| 70 °C        | PP / HDPE                       | ~0.4×               |                                               |
| 60 °C        | PVC (non-pressurized tray only) | ~0.1–0.17×          | Degraded mode: holds mothball / 2-person duty |

**Air-swept backup (degraded mode, X8-corrected):** small EC fan, PP demister, and — per X8 rule 3 — a **raw-water condenser** (DHW-preheat first stage) on the ~98 g/kg / 53 °C-dew-point exhaust instead of a bare overboard duct: ~64 g/kg recoverable at a 34 °C approach. Condensate is **technical-grade only, PENDING TDS assay** (never M-cycle feed or potable before it clears). Any residual duct run still slopes continuously down and out with a drip leg — the exhaust rains salty condensate on anything cooler than 53 °C.

**Crystallization interlock (safety register item 6):** 40 wt% liquidus  $\approx 12$ –13 °C; 44 wt%  $\approx 22$  °C. **High-SG cutoff at 42 wt% unless brine  $\geq 22$  °C guaranteed; stored concentrate capped at 43 wt%.** The tropical risk is *over-concentrating* on a strong-heat day.

## 5. Ventilation hardware — ERV and the CO<sub>2</sub> battery (X7/X10)

- **ERV core:** polymer-membrane counterflow enthalpy exchanger,  $\epsilon_{lat} \geq 0.8$  (effectiveness is purchasable with membrane area — oversize it), CO<sub>2</sub> crossover / EATR <5% (test E). Fresh side upstream of the absorber; exhaust side fed by the **ducted exhaust network** (pickups high in every closable room + galley hood — the one architecture change from diffuse positive-pressure exfiltration to semi-balanced flow). PENDING test E: real  $\epsilon_{lat}$ , salt-aerosol fouling trend (U-tube  $\Delta P$  gauge), condensation behavior at DP-A inlet.
- **CO<sub>2</sub> battery:** two-bed solid-amine TSA per doc 00 §5 (bed chemistry per the X11 heat-grade ladder — the K<sub>2</sub>CO<sub>3</sub>/carbon variant requires a  $\geq \sim 130$  °C tap and is waste-heat-platform hardware, never solar-bus hardware; the solar path keeps the amine bed unchanged), in the recirculation branch post-absorber (45–55% RH), chloride-free side → ordinary materials. Regen coil from the buffer bus / pre-temper branch at 85–95 °C; CO<sub>2</sub>-rich purge ducted out. **Required-PENDING test J** (slip assay gates cabin connection — safety register item 4).
- **Sensors/interlocks:** per-room NDIR on the ESP32 bus, interlock on the max; fresh damper with mechanical minimum stop (doc 00 §5/§8).

## 6. Brine circulation and storage

- **Transfer pump:** peristaltic (Norprene/santoprene tube) — brine touches only the tube, inherently metered, self-priming, a few watts. Recirculation: PP mag-drive, ceramic spindle, EPDM/FKM seals.
- **Tubing:** PE/PEX with plastic push-fits (PP/POM bodies) on the cool side; **CPVC solvent weld or PP compression on hot runs**. No brass-body push-fits.
- **Valves:** true-union PVC/CPVC ball, EPDM/FKM seats.
- **Reserve:** 25–60 L vented HDPE tanks, tied down (DP-A occupied-night draw  $\sim 35\text{--}40$  kg — doc 10 §3). Drip containment tray under everything: concentrated CaCl<sub>2</sub> drips never dry. Every wall penetration gets a drip loop.

## 7. Thermal bus (chloride-free side)

A **30–60 L insulated buffer tank (65–90 °C)** merges all heat sources, kills short-cycling, and doubles as DHW and space heat.

| Source                             | Coupling                                                              | Notes                                                                                                                   |
|------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Diesel hydronic heater, 5 kW class | Closed glycol loop → plate HX → buffer                                | $\sim 1.1\text{--}1.4$ L/day heater-only full DP-A duty (with ERV); marine: sealed exhaust, swan-neck, own standpipe    |
| Solar, top lift                    | <b>U-pipe evacuated tubes, 2.5–4.5 m<sup>2</sup></b> direct to buffer | U-pipe, not heat-pipe — works flush-horizontal (marine roll / flat roofs). Array sized to the 9–11 kWh/day ERV'd budget |
| Solar, preheat (optional)          | PVT hybrid → buffer bottom                                            | $\sim 55$ °C max; carries $\sim 60\%$ of enthalpy while making PV watts                                                 |
| Engine jacket / site waste heat    | Second coil of twin-coil calorifier or second Ti HX                   | Regeneration free underway / co-located with process heat                                                               |

Buffer → regenerator: 12 V circulator (5–15 W) through a **thermostatic limit valve (85–90 °C)** into a **titanium tube-in-shell HX, brine through the Ti tubes**. Zero-metal fallbacks: long CPVC coil in the pool ( $\sim 3\text{--}5\times$  area) or a bain-marie. Controls: aquastat band; regen transfer on buffer >60 °C AND brine SG < setpoint; low-level cutoffs; CO<sub>2</sub>-bed regen valve slaved to buffer temperature (solar-window scheduling). One ESP32 with logging covers all of it.

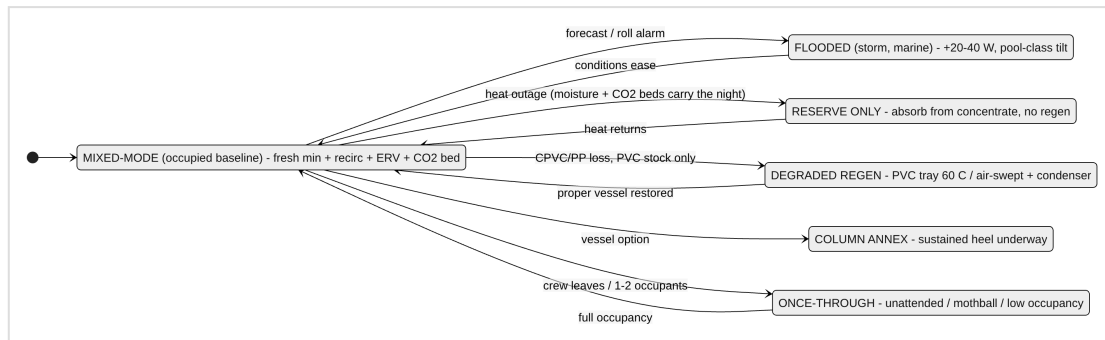
## 8. Raw-water circuit

The largest single electrical consumer at scale ( $\sim 30\text{--}50$  W of the 40–80 W total). One consolidated circuit + manifold (Ti/plastic wetted only) serves: absorber sump coil (**required for comfort** — 0.66 kW at 2 K rise  $\approx 300$  L/h), still condenser, air-swept-mode condenser, and M-cycle rejection. Run 3–4 K rise to halve flow (costs  $\sim 1$  g/kg of floor). Marine: scoop pickup underway; land: lake/well loop, or an evaporative fluid cooler / dry cooler with the doc 00 §1 penalties stated.

## 9. Bubble-column annex (sustained-heel marine; otherwise optional)

For sustained 15–20° heel, pools with rising bubbles beat any film media. 4 × 6" clear PVC absorber columns (port/starboard pairs on a common manifold — heel self-averages), 1 × CPVC regen column, 10–15 cm pools. Hard-won specifics: airflow regime limit 0.2 m/s superficial → 13.4 m<sup>3</sup>/h per column; sparger head **1.37 kPa per 10 cm** submergence at SG 1.4 *plus* diffuser dynamic wet pressure — **EPDM membrane diffusers add 1.5–4 kPa DWP** (rising with fouling) and silently double or triple blower watts; **drilled CPVC rings (1 mm holes,  $\sim 0.3\text{--}0.5$  kPa) win unless measured approach fraction says otherwise**. Air check valve + mounting above max liquid level + anti-siphon loop on every air line. Passive submergence via internal overflow standpipes. Electrical  $\sim 100\text{--}140$  W (drilled rings) vs 20–40 W film — the reason columns are the annex.

## 10. Operating and degradation modes



CO<sub>2</sub> interlock active in every occupied state; ventilation floor guaranteed by the damper's mechanical stop even in Battery and Degraded states.

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### Version history

- **v1.0** — New lineage from archived core-02: ERV and CO<sub>2</sub>-battery hardware integrated (X7/X10); air-swept mode gains the X8 condenser; mixed-mode state added; raw-water sink generalization; brazed-plate prohibition and materials law carried verbatim.
- **v1.1** — §5 CO<sub>2</sub>-battery bullet gains the X11 pointer (carbonate variant is waste-heat-platform hardware only; the liquid solar path is unchanged).
- **v1.2** — §3 gains the two-duties note separating outlet-side brine mist elimination (test B, unchanged) from intake-side sea-salt removal (doc 22 §3), with the two-worlds law restated as never relaxed by intake filtration — the desiccant is itself the chloride source regardless of site. No figure changed.